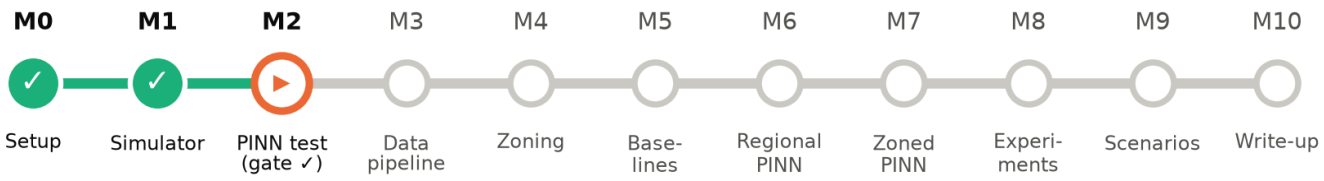
# Zoned PINNs for Groundwater Forecasting

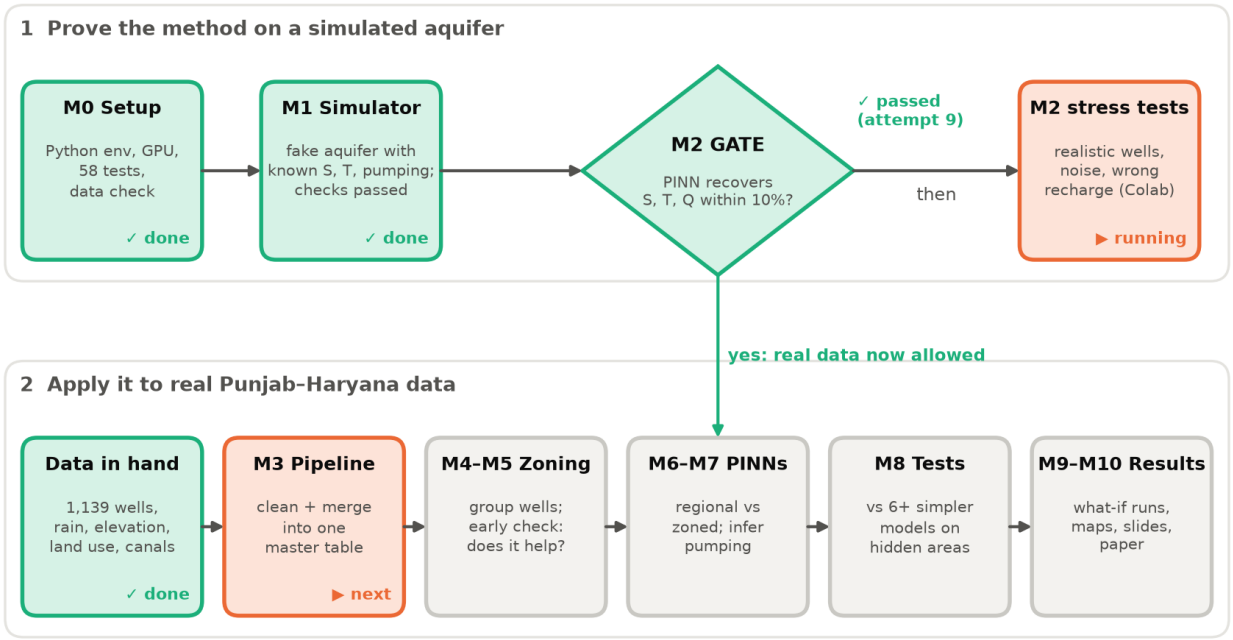
Progress report · 21 October 2026 · Punjab–Haryana belt, 16 districts · **M0–M1 done · M2 gate passed**




## The project in one paragraph

We forecast groundwater levels everywhere in the study area, including places with no monitoring well. Wells are grouped into **zones** that behave alike. Each zone gets a **physics-informed neural network (PINN)**: it fits the measured water levels while obeying the groundwater flow equation **$S \cdot \partial h/\partial t = T\nabla^2 h + R - Q$** (storage × change in level = sideways flow + recharge − pumping). Unmetered farm **pumping Q** is inferred as a hidden quantity. Before touching real data, the method must prove on a simulated aquifer that it recovers answers we planted.

## Project flow and where we are




## Done so far

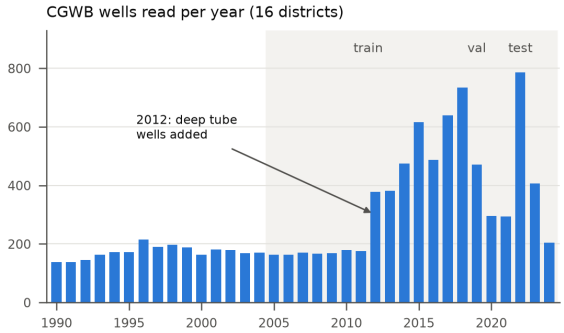
| Step | What was done | Key result |
|---|---|---|
| Plan | Brief checked; 12 design problems ranked and fixed (PLAN.md) | Pumping made identifiable by tying it to land use and the crop calendar |
| Data | Study area chosen by a data check; every dataset fetched by script | 1,139 wells (1990–2024) + 248 sensor wells; nothing left to download by hand |
| M0 Setup | Python environment, project layout, automatic tests, Colab notebook, literature check | GPU (GTX 1050) works; 58 tests pass; no prior zoned or hidden-pumping PINN on field data found |
| M1 Simulator | Finite-difference aquifer simulator and 4 synthetic aquifers (A–D) with known answers | Matches exact textbook solutions; water balance closes to 1 part in 10 billion |
| M2 PINN test | The real PINN trained on a simulated aquifer from deliberately wrong starting values; 9 attempts, each failure diagnosed and fixed (page 3) | **Gate passed:** storage −5%, flow ease +2%, total pumping −1%; stress tests running on Colab |

## Biggest risks found in the plan review, and how they are handled

| Risk | Why it matters | How the plan handles it |
|---|---|---|
| Pumping can't be pinned down | Many mixes of storage, recharge and pumping fit the same water levels | Pumping tied to land use and crop season; tested on simulated data first; reported as ranges |
| Physics may add little | Wells are 5–10 km apart, but water moves only ~1 km sideways per season | A simple 'bucket' model and a no-sideways-flow version are tested, so we can say so honestly |
| Result could be luck | A small test set makes a 5% difference look like chance | 859 test wells; 3 different held-out area layouts; confidence ranges; main result fixed in advance |
| PINN training fails | PINNs often fail to train on messy, sparse data | **Happened in M2 attempt 1** (physics under-weighted); diagnosed and fixed, see page 3 |

## Data in hand

| Dataset | Coverage |
|---|---|
| CGWB water levels (India-WRIS) | 1,139 wells, 1990–2024, with well type and depth |
| Sensor wells (6-hourly) | 248 wells, mostly 2022–2025 |
| IMD rainfall (0.25°, daily) | All 35 years, 1990–2024 |
| Elevation (Copernicus 30 m) | Whole belt |
| Land use (Bhuvan, ESA) | 2005-06, 2011-12, 2015-16, 2021 |
| Canals, districts, crop areas | OSM canals; crop areas 1997–2025 |
| CGWB reports (PDF) | Aquifer maps for all 16 districts; assessments 2020–24 |

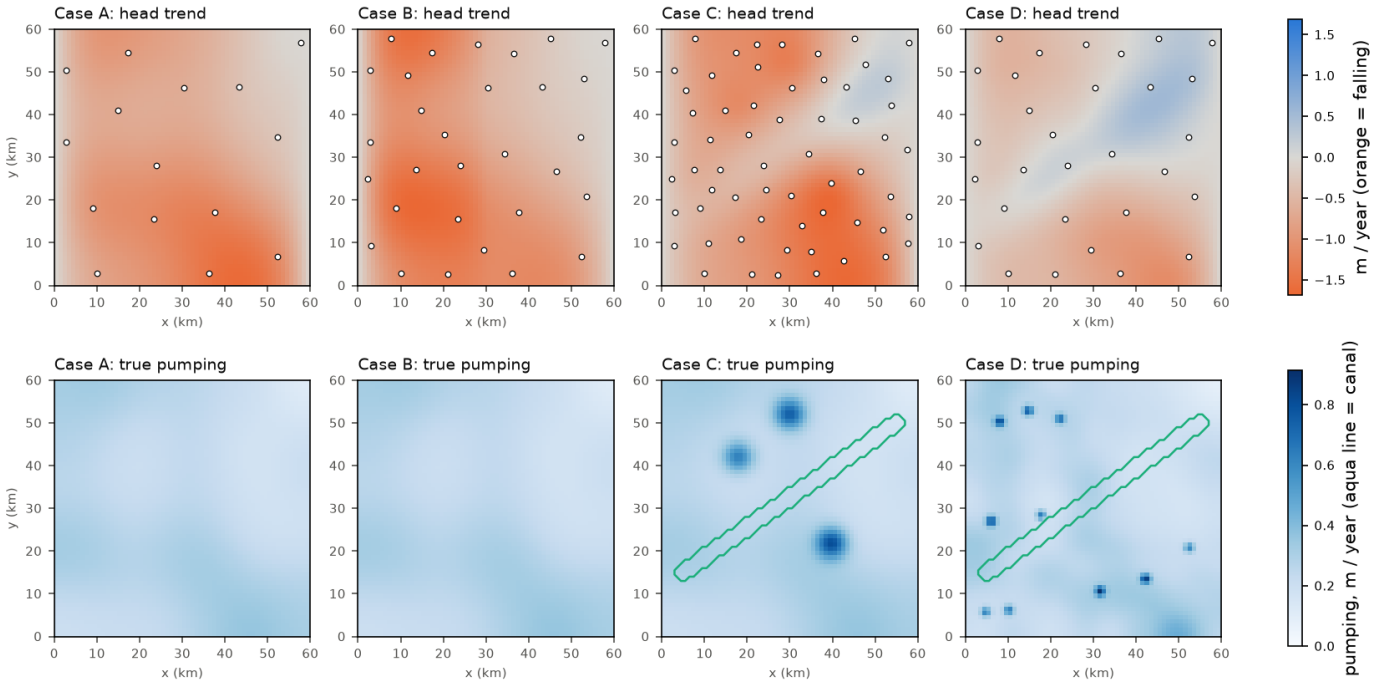



**Usable for the main analysis:** 1,031 wells (shallow, unconfined aquifer) · **193** with long enough records to define zones · **859** wells in the 2020–24 test years (663 with earlier history, 196 brand new: a fair test of predicting where the model has never seen data).
**Watch-outs:** the network shifted from shallow dug wells to deeper tube wells in 2012 (check that zones are not just well types); Karnal is thin in WRIS (filled from the India Data Portal); land use exists as snapshots, not every year.

## M1 result: the simulated aquifer is verified

| Check against an exact answer | Required | Result |
|---|---|---|
| Grid made 2× finer → error shrinks | ≈ 4× | 3.9× ✓ |
| Water conserved every simulated day | < 1 in $10^8$ | 1 in $10^{10}$ ✓ |
| Closed box / straight-line cases | exact | exact ✓ |
| Single pumping well (Theis) | close | within 0.4% ✓ |

**Four test aquifers** (60 × 60 km, 10 years):
**A** uniform, falls ~0.9 m/yr: main test
**B** two zones with different S, T: can zoning find them?
**C** canal + 3 heavy-pumping spots (4–7 m dips): **the M2 pass/fail test**
**D** deliberately unfair: hidden recharge, patchy pumping, blurry land-use map: measures how wrong the model gets when its assumptions are wrong

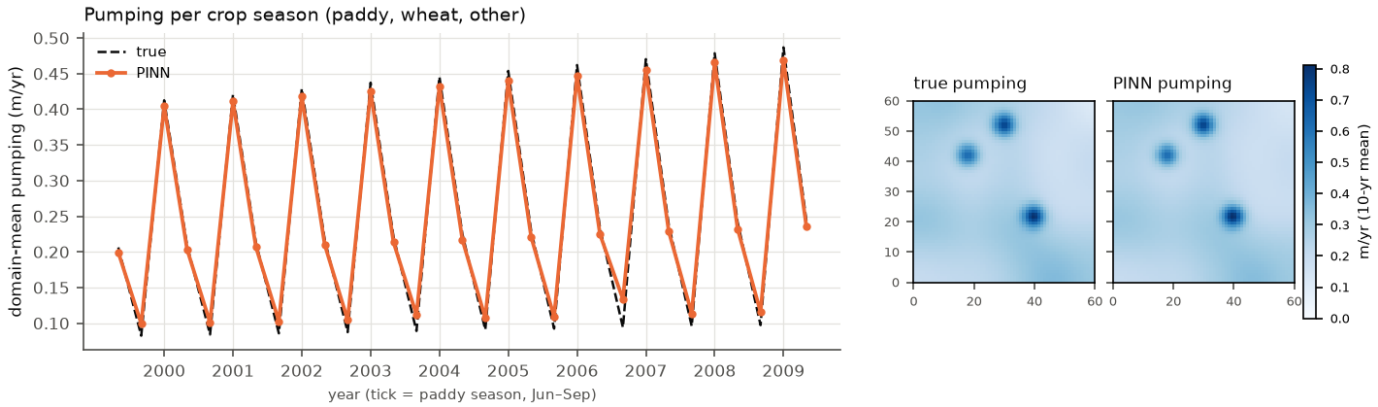



Top: 10-year water-level trend in each test aquifer (orange = falling; dots = the sparse 'wells' the PINN will see). Bottom: the true pumping the PINN must recover without being shown it (aqua outline = canal).

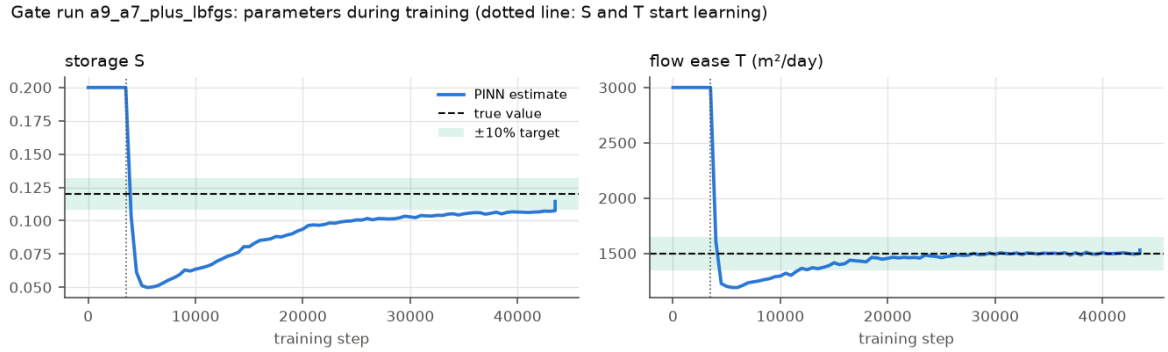
## M2 result: the PINN recovers a hidden aquifer (gate passed)

The real PINN was trained on simulated case C (60 wells read monthly). It started from wrong guesses: storage 0.20 instead of 0.12, ease of flow 3,000 instead of 1,500 m²/day. It got no hints. To pass, storage, ease of flow and pumping per season all had to come out within 10% of the hidden truth. ✓ marks a value within 10%.

| Attempt | What changed | Storage S | Flow ease T | Pumping per season | Total pumping |
|---|---|---|---|---|---|
| 1 | Automatic balancing of data vs physics | -48% | -30% | 25% | -20% |
| 5 | Fixed balance; month-of-year pumping input | -2% ✓ | -2% ✓ | 59% | -0% ✓ |
| 6 | + pumping blurred like recharge in the physics check | -19% | -3% ✓ | 15% | -7% ✓ |
| 7 | + gentle start for S, T; twice the training | -10% | -0% ✓ | 14% | -4% ✓ |
| 9 | + final L-BFGS polish (passed) | -5% ✓ | +2% ✓ | 9% ✓ | -1% ✓ |




Attempt 9: pumping per crop season over 10 years, PINN vs truth (left), and the 10-year average pumping map (right). The PINN was never shown either.




How storage S and ease of flow T moved during training of attempt 9 (green band = within 10% of the truth).

## What went wrong on the way, and the fixes (all kept as project defaults)

| Problem found | Fix |
|---|---|
| Automatic loss balancing let the data swamp the physics | Fixed weight between data and physics |
| Smooth seasonal pumping made storage come out too low | Month-of-year input for pumping (the crop calendar) |
| Blurred recharge vs sharp pumping let the model trade one for the other | Pumping blurred over the same 31 days inside the physics check |
| Storage jumped too low when it started learning, then crawled back | Gentle start for S and T, twice the training, final L-BFGS polish |

## Caveats, next steps and honest status

| | |
|---|---|
| **Caveats** | Shoulder months (April, May, October) are overestimated by about 22% (paddy and wheat seasons are within 4%). One training run only; a repeat with a different random start is running. This was the friendly case: realistic sparse, noisy data will do worse, and will be reported, not gated. |
| **Now (Colab)** | M2 stress tests: 30 wells read 4×/year, noise, recharge 20% wrong, the unfair case D, and the brief's original pumping design. |
| **Next** | M3 data pipeline (real data) → M4–M5 zoning and the early 'does zoning help?' check → M6 first PINN on real data. |
| **Honest status** | No accuracy result on real data exists yet. The 5–10% improvement over a single regional PINN is a *target, not a result*; first evidence comes from the M4 check. Pumping on real data will be an inference without ground truth, reported as ranges. |